

Short-term synaptic plasticity

Computational Neuroscience

University of Bristol

Adapted from B Mizusaki's slides

M Rule

Learning outcomes:

- ▶ Describe the concept, mechanisms of Short-Term Plasticity (STP)
- ▶ Be able to **write down** the equations for a simple model of STP
- ▶ Be able to describe STP as an information filter.

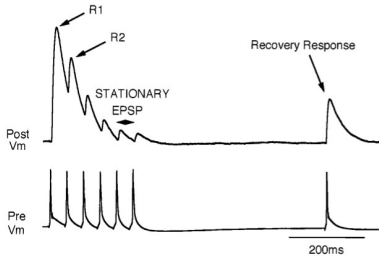
Short-term plasticity

Short-term synaptic plasticity is thought to be involved in fast cognitive processing.

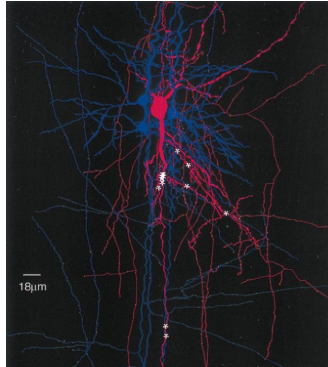
- ▶ Similar timescale to electrical dynamics
 - Short-term synaptic facilitation (strength increases with use)
 - Short-term synaptic depression (strength decreases with use)

STP in an experiment

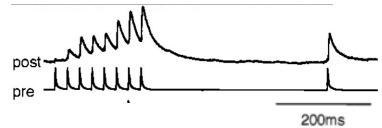
Depression



Tsodyks & Markram, PNAS (1997)



Facilitation



Markram, Wang, & Tsodyks, PNAS (1998)

Short-term depression

Too many vesicles are used before more of them are prepared to be released

- ▶ Depletion of the readily releasable pool

Rate of vesicle replenishment limits rate of transmission:

- ▶ Low-pass filter

Draw what happens with synaptic vesicles and post-synaptic potentials

Short-term facilitation

AP induces local Ca^{2+} influx in the *presynaptic* terminal

- ▶ Elevated Ca^{2+} leads to increased release probability
- ▶ Ca^{2+} trigger for vesicle fusion nonlinear (release still requires presynaptic spike)

Time of Ca^{2+} availability in the terminal limits period of available facilitation:

- ▶ High-pass filter

Draw what happens with synaptic vesicles and post-synaptic potentials

Short-term depression model

Think of the synaptic efficacy as being determined by the available amount of a resource

$$\begin{aligned}
 \dot{R} &= \rho_{\text{rec}} I - D \cdot R \cdot \delta(t - t_s) \\
 \dot{E} &= -\rho_{\text{inac}} E + D \cdot R \cdot \delta(t - t_s) \\
 I &= 1 - R - E
 \end{aligned}$$

$R \in \{0, 1\}$ fraction “readily” available

- ▶ Decrements by dR on each spike
- ▶ Recharges from I fraction rate $\rho_{\text{rec}} I$
- ▶ $D \in [0, 1] \sim$ portion resource depleted with each spike

$E \in \{0, 1\}$ fraction “exhausted”:

- ▶ Increments by $D \cdot R$ on each spike
- ▶ $E \rightarrow I$ rate $\rho_{\text{inac}} E$

$I \in \{0, 1\}$ fraction “inactive”

- ▶ Recycled, but not ready to use
- ▶ $I \rightarrow R$ at rate $\rho_{\text{rec}} I$

The lecturer probably needs to explain that $\delta(t - t_s)$ notation

The Dirac delta notation $q \cdot \delta(t - t_s)$ on the RHS of an ODE is shorthand for a jump condition

- ▶ It means that we should add amount q to the integrated variable when we pass through time t_s
- ▶ You may also see notation t^- and t^+ to denote that two values may be associated with a single time point before and after the jump.
- ▶ It is tempting to assume $\delta(t - t_s)$ can always be viewed as an impulse as in the case of linear ODEs
 - Not always true when the condition generating the impulse comes from the ODE's own state.
 - Writing out the jump conditions explicitly might be more clear.

But, neuroscientists love writing spike (or event) sequences as impulse trains $y(t) = \sum_{t_s \in \mathcal{T}} \delta(t - t_s)$.

How should the (R, E, I) modulate synaptic weight?

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Here is a first-order exponential decay synapse ($q \sim$ neurotransmitter released)

$$\begin{aligned}g_s(t) &= \bar{g} \cdot s(t) \\ \tau_s \dot{s} &= -s + q \cdot \delta(t - t_s)\end{aligned}\tag{1}$$

We could do this

$$\begin{aligned}g_s(t) &= \bar{g} \cdot s(t) \\ \tau_s \dot{s} &= -s + R \cdot q \cdot \delta(t - t_s)\end{aligned}\tag{2}$$

Can you think of other reasonable choices?

What would we change to get a facilitatory model?

Summary

STP can be modelled as a competition for resources

STP works as a frequency filter

- ▶ STF: Low pass
- ▶ STD: High pass

Most neurons have synapses that are either mostly facilitating or depressing

end

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin-Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch-Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative